

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Technology Stack

Online Complaint Registration and Management Systems — MERN
Stack E-commerce Application

Phase: Requirement Analysis Phase

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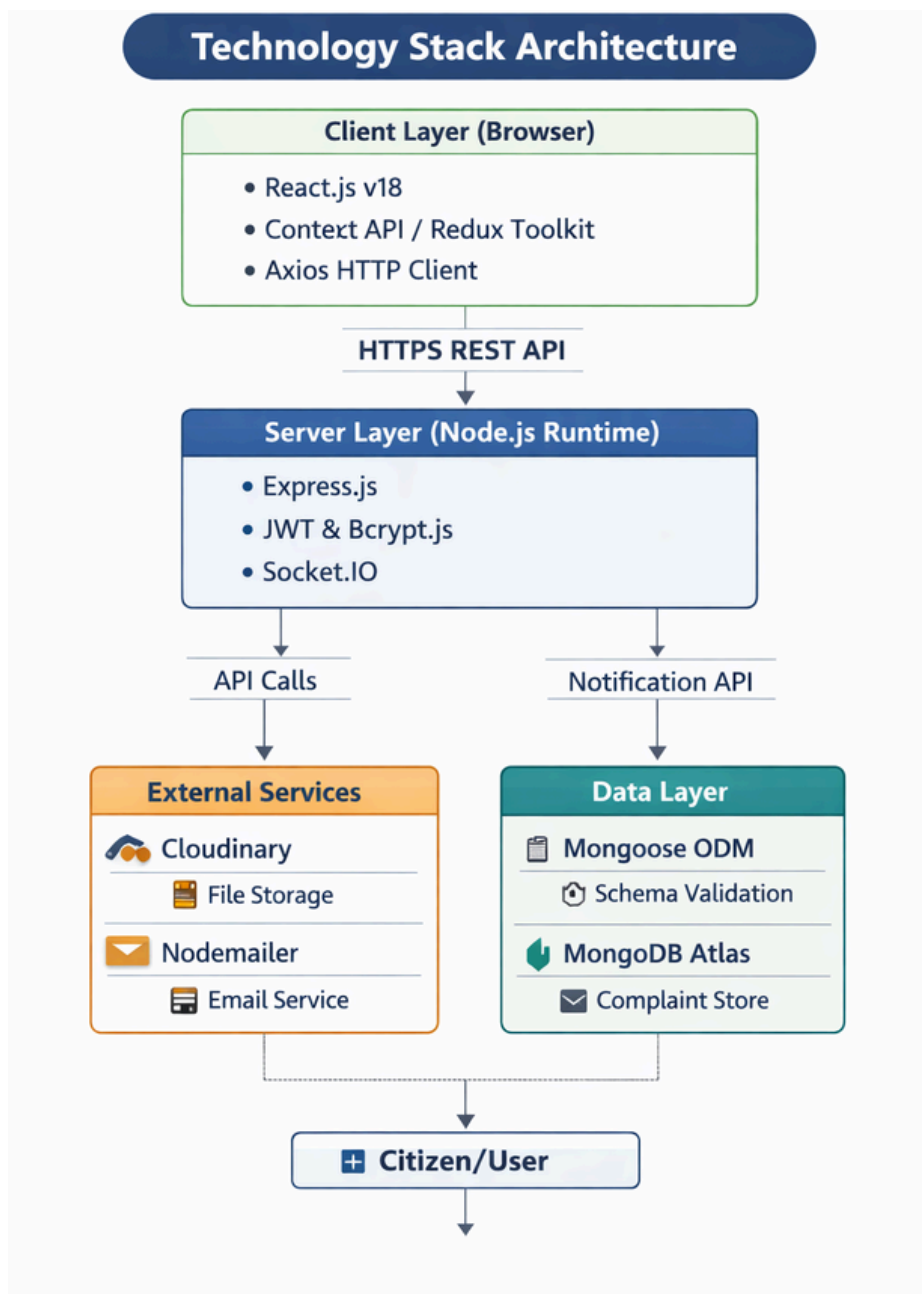
Classification: Project Documentation — For Academic Submission

Technology Stack

1. Introduction

This document provides a formal specification of the technology stack selected for the Online Complaint Registration and Management System (OCRMS). Each technology selection is accompanied by its architectural role, justification, and a brief Architecture Decision Record (ADR) summarising the decision rationale and alternatives considered

2. Integration Architecture Diagram



3. Detailed Stack Breakdown

Layer	Technology	Version	Purpose & Justification	
Frontend Framework	React.js	v18.x	Component-based SPA architecture enabling responsive and dynamic user interfaces for complaint management.	
State Management	Context API / Redux Toolkit	v2.x	Centralised global state for authentication context, shopping cart, and cached API data. Redux Toolkit reduces boilerplate via	
HTTP Client	Axios	v1.x	Handles communication with request/response interceptors and automatic JSON serialization.	API with
Backend Runtime	Node.js	v18 LTS	Event-driven JavaScript runtime supporting scalable API request handling.	
Backend Framework	Express.js	v4.x	Lightweight framework for building	

Layer	Technology	Version	Purpose & Justification
Authentication	JSON Web Tokens (JWT)	RFC 7519	RESTful APIs, middleware management, and routing.
			Stateless authentication mechanism stored in HTTP-only cookies for enhanced security
Password Security	Bcrypt.js	v5.x	Secure password hashing with salt rounds ≥ 10 before database storage.
Database	MongoDB (Atlas)	v6.x	Cloud-hosted NoSQL database for storing users, complaints, feedback, and notification records
ODM	Mongoose	v8.x	Schema-based ODM providing validation, relationships, middleware hooks, and data integrity

Layer	Technology	Version	Purpose & Justification
Real-Time Communication	Socket.IO	Latest	Enables real-time complaint updates, notifications, and user-officer chat functionality.
Media Hosting	Cloudinary	v2.x SDK	Stores complaint evidence images and documents with CDN optimization.
Deployment	Vercel (FE) / Render (BE)	—	Provides scalable frontend hosting and backend deployment with automatic CI/CD integration

4. Architecture Decision Records (ADRs)

ADR-01: MongoDB over Relational Database

Field	Detail
Status	✓ Accepted
Context	Complaint records, feedback, notifications, and user information require flexible document structures
Decision	Adopt MongoDB as the primary data store.
Justification	Document-oriented storage supports evolving complaint categories and dynamic complaint data without extensive schema migrations.
Consequences	Complex relational queries require aggregation pipelines. Mitigated using Mongoose schemas and indexing strategies.

ADR-02: JWT in HTTP-only Cookies over localStorage

Field	Detail
Status	✓ Accepted
Decision	Store JWT in HTTP-only, SameSite=Strict cookies rather than <code>localStorage</code> .
Justification	Prevents token exposure through client-side scripts and reduces XSS-related risks.
Consequences	Requires CORS configuration and CSRF protection mechanisms.

ADR-03: Cloudinary over Local File Storage

Field	Detail
Status	✓ Accepted
Decision	Use Socket.IO for real-time complaint status updates and communication.
Justification	Enables instant notifications between citizens, officers, and administrators without continuous polling.
Consequences	Requires persistent WebSocket connections and additional server resource management

5. Conclusion

The selected technology stack provides a secure, scalable, and efficient foundation for OCRMS. By leveraging React, Node.js, Express, MongoDB, JWT authentication, and Socket.IO, the system ensures reliable complaint registration, tracking, communication, and resolution while supporting future scalability and maintainability.